STAT 519/619 Project

February 12, 2026

1 Important deadlines and due dates

- Group registration: Feb 29th.
- Proposal is due on: March 14th.
- Presentation: In-class during last 2 weeks.
- Written report: April 14, 11:59pm.

2 Project

The project is to carry out Bayesian analysis on a chosen data set. The entire project includes the proposal, presentation, and final written report. You may work on this project with a group of 1-2 students. Send me an email before Feb 29th to confirm your group members.

2.1 Proposal

The project proposal is due on March 14th. In the proposal, you are required to justify your choice of a data set, formulate a research question, and specify potential Bayesian modeling approaches with the sampling/optimization techniques (at least 2 different MC methods). I will confirm your proposal or revise it if it does not fit the scope of the course.

2.2 Presentation

For each group, schedule a day from any of April 2,4,9.

For groups of 1 student, the presentation will be 10 minutes. For groups of 2 students, the presentation will be 15 minutes and both students are required to present (split the presentation equally between 2). The presentation should include at least the following parts:

- Introduction
- Methods
- Results
- Discussions

Each group should expect 2-3 questions from the instructor or the audience following their presentation. Usage of interactive/animated tools and plots is highly encouraged.

2.3 Written report

The final written report includes at least the following sections:

- Introduction: data background, existing methods for analyzing such data, motivation for doing Bayesian statistics, and formulating a research question. Illustrative graphs are a great way of introducing your project.
- Model specification: choice of models; choice of priors. And justify your model. For example, why do you choose certain priors with fixed hyper-parameters? Why put a hyper-prior on priors? For example in regression models, why do you choose to include a set of covariates?
- Data analysis: posterior inference, posterior sampling, and interpretation of application results. For sampling the posterior, you should include at least two different methods from importance sampling, MCMC, variational methods, or more.
- Model checking: how different prior distributions affect the model outcome. Prior/Posterior checks. Sampling diagnostics (effective sample size, convergence, etc.)

- Summary and conclusion.
- Codes should be submitted in a separate file along with your project. I should be able to reproduce graphs and results in your project.

The written report should not exceed 15 pages in length.

3 Data

You may choose a data set from

- R datasets: https://vincentarelbundock.github.io/Rdatasets/datasets.html
- Kaggle competition
- Data sets from papers published in Journal of the American Statistical Association; Journal of the Royal Statistical Society Series B; Statistics and Computing; Journal of Computational and Graphical Statistics; Biometrics; Journal of Machine Learning Research.

4 Miscellaneous

- You may use AI to help you finish the project. However, **you need to record the EXACT prompts used for generating texts or codes. You need to clearly indicate WHERE you used AI**. Failing to do so will be considered a violation of academic integrity. All suspected plagiarism will be recorded and investigated.
- Clear references. You may seek help with other works, but you need to have the proper references in your proposal/presentation/report. You have to justify how others work to assist you in doing your own analysis.